Nicole Esther Lee

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Game programmer and computer science student who has built engine systems in C++ and a 3D level in Unreal Engine 5.

EDUCATION

Singapore Institute of Technology – DigiPen Institute of Technology Singapore
Bachelor of Science in Computer Science in Interactive Media and Game Development *Aug 2023 – 2027 (expected)*

Millennia Institute Singapore
GCE A-Levels *Feb 2020 – Nov 2022*

GAME PROJECTS

Friends Frog-ever (Ribbit Express) – 3D Game Engine for Android and Windows Aug 2025 – Apr 2026
C++ · Level Editor Lead

- Built a Unity-style level editor with dynamic component panels that add and remove components at runtime
- Built the scene viewport with ImGuizmo for direct translate, rotate and scale of objects
- Structured the editor as modular systems for a real-time 3D editing workflow
- Led the level editor, turning design students' workflow requests into features with the other engine programmers

Chroma – 2D Platformer Game Engine Aug 2024 – Apr 2025
C++ · Team Lead (8 people)

- Built a physics system from scratch with gravity and collision detection
- Designed the core input system for player movement and level-object interactions
- Programmed gameplay systems: colour-changing platforms and dynamic player movement
- Built level editor tools: asset browser, entity/component panels and an interactive viewport

Nyx: Dreams of the Stray – 3D Narrative Level Jun 2025
Unreal Engine 5, Blueprints · Grade A

- Built a six-scene level in Blueprints, shipped as a playable Windows build: a guide character that vanishes as you approach, collectibles, a countdown timer, roaming NPCs, respawns and scene transitions
- Ran 6 playtests across 3 builds; time played rose from 3:09 to 7:48 and platforming falls dropped from 5 to 0 after I retuned jumps and added respawns

EXPERIENCE

DBS Bank Singapore
Site Reliability Engineer Intern, Global Financial Markets Technology *May 2026 – Present*

- Automate routine operations with Bash/shell scripts and Power Automate, including scheduled report runs, job schedules and email alerts
- Debug issues across Linux and AWS environments and analyse logs with the ELK stack

SKILLS

Languages: C++, C, JavaScript, Bash/Shell

Engines & Graphics: Custom C++ engines, Unreal Engine 5 (Blueprints), EnTT (ECS), physics, level editors, OpenGL/GLSL, ImGui/ImGuizmo

Tools: Git/GitHub, Linux, Jira